Exponentials & Logarithms

ClipSAT Printable Practice Suite (clipsat.org) • Topic 8.01

ANSWER KEY — COMPLETE WORKED SOLUTIONS

Part I — Free-Response Questions (FRQs): Foundational Fluency (1–7)

1 **[Foundational]** • **Target: Exponential growth and decay models**

The population of a town is modeled by $P(t) = 5000e^{0.03t}$, where t is in years. Find $P(10)$, rounded to the nearest whole number.

$P(10) = 5000e^{0.3} = 5000(1.34986) \approx 6749.$

2 **[Foundational]** • **Target: Evaluating exponential expressions**

Evaluate 2^5.

$2^5 = 32.$

3 **[Foundational]** • **Target: Evaluating exponential expressions**

Evaluate 3^{-2}.

$3^{-2} = \frac{1}{9}.$

4 **[Foundational]** • **Target: Evaluating exponential expressions**

Evaluate $16^{1/2}$.

$16^{1/2} = 4.$

5 **[Foundational]** • **Target: Evaluating exponential expressions**

Evaluate $27^{2/3}$.

$27^{2/3} = (27^{1/3})^2 = 3^2 = 9.$

6 **[Foundational]** • **Target: Evaluating logarithms**

Evaluate $\log_2 32$.

Since $2^5 = 32$, $\log_2 32 = 5$.

7 **[Foundational]** • **Target: Evaluating logarithms**

Evaluate $\log_3 81$.

Since $3^4 = 81$, $\log_3 81 = 4$.

Part II — Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

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8 [Intermediate] • Target: Evaluating logarithms

Evaluate $\log_5 1$.

$\log_5 1 = 0$.

9 [Intermediate] • Target: Evaluating logarithms

Evaluate $\log_{10} 0.01$.

Since $10^{-2} = 0.01$, $\log_{10} 0.01 = -2$.

10 [Intermediate] • Target: Log laws: product and quotient

Write $\log_2 (8x)$ as a sum of logarithms.

$\log_2 (8x) = \log_2 8 + \log_2 x = 3 + \log_2 x$.

11 [Intermediate] • Target: Log laws: product and quotient

Write $\log_3 \left(\frac{x}{9}\right)$ as a difference of logarithms.

$\log_3 \left(\frac{x}{9}\right) = \log_3 x - \log_3 9 = \log_3 x - 2$.

12 [Intermediate] • Target: Log laws: product and quotient

Combine $\log_5 5 + \log_5 25$ into a single logarithm and evaluate.

$\log_5 (5 \times 25) = \log_5 125 = 3$ (since $5^3 = 125$).

13 [Intermediate] • Target: Log laws: product and quotient

Combine $\log_2 24 - \log_2 3$ into a single logarithm and evaluate.

$\log_2 \left(\frac{24}{3}\right) = \log_2 8 = 3$.

14 [Intermediate] • Target: Log laws: the power rule

Write $\log(x^5)$ using the power rule.

$\log(x^5) = 5\log x$.

Part III — Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

15 [Challenging] • Target: Log laws: the power rule • Cross-Lesson: 6.01 Limits & Derivatives

Evaluate $\log_2 (4^3)$ using the power rule.

$\log_2 (4^3) = 3\log_2 4 = 3(2) = 6$.

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16 [Challenging] • Target: Log laws: the power rule • Cross-Lesson: 7.01 Integration & Applications

Write $2\log_3 x + \log_3 y$ as a single logarithm.

$2\log_3 x + \log_3 y = \log_3 (x^2 y)$.

17 [Challenging] • Target: Expanding and condensing logarithmic expressions • Cross-Lesson: 9.01 Statistics & Probability

Expand $\log\left(\frac{x^3 y}{z^2}\right)$ fully.

$3\log x + \log y - 2\log z$.

18 [Challenging] • Target: Expanding and condensing logarithmic expressions • Cross-Lesson: 1.01 Algebra & Equations

Condense $3\ln x - 2\ln y$ into a single logarithm.

$\ln\left(\frac{x^3}{y^2}\right)$.

19 [Challenging] • Target: Expanding and condensing logarithmic expressions • Cross-Lesson: 2.01 Functions & Transformations

Expand $\log_2\left(\sqrt{x} \cdot y^3\right)$ fully.

$\sqrt{x} = x^{1/2}$, so the expression equals $\frac{1}{2}\log_2 x + 3\log_2 y$.

20 [Challenging] • Target: Solving exponential equations (matching bases) • Cross-Lesson: 3.01 Sequences & Series

Solve $2^x = 32$.

$32 = 2^5$, so $x = 5$.

Part IV — Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

21 [Foundational] • Target: MCQ — Exponential growth and decay models

The graph shows an exponential function. Based on the graph, does it represent growth or decay?

a. Neither

b. **Growth ✓**

c. Cannot be determined

d. Decay

The curve rises as x increases: growth.

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22 [Foundational] • Target: MCQ — Evaluating exponential expressions

Evaluate 3^4.

a. 81 ✓

b. 243

c. 12

d. 64

$3^4 = 81$.

23 [Foundational] • Target: MCQ — Evaluating exponential expressions

Evaluate $8^{2/3}$.

a. 5.33

b. 4 ✓

c. 2

d. 16

$8^{2/3} = (8^{1/3})^2 = 2^2 = 4$.

24 [Foundational] • Target: MCQ — Evaluating logarithms

Evaluate $\log_2 64$.

a. 32

b. 8

c. 5

d. 6 ✓

$2^6 = 64$.

25 [Foundational] • Target: MCQ — Evaluating logarithms

Evaluate $\log_5 125$.

a. 5

b. 3 ✓

c. 1

d. 25

$5^3 = 125$.

26 [Foundational] • Target: MCQ — Log laws: product and quotient

Combine $\log_3 54 - \log_3 2$ into a single logarithm and evaluate.

a. 27

b. 3 ✓

c. 26

d. 52

$\log_3 (54 \div 2) = \log_3 27 = 3$.

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27 [Foundational] • Target: MCQ — Log laws: the power rule

Write $4\log x - \log y$ as a single logarithm.

a. $\log\left(\frac{x^4}{y}\right)$ ✓

b. $\log(4x - y)$

c. $4\log\left(\frac{x}{y}\right)$

d. $\log\left(\frac{4x}{y}\right)$

$4\log x = \log(x^4)$; combined: $\log\left(\frac{x^4}{y}\right)$.

28 [Foundational] • Target: MCQ — Expanding and condensing logarithmic expressions

Condense $\frac{1}{2}\log x + 2\log y$ into a single logarithm.

a. $\log(x \cdot y^2)$

b. $\log\left(\frac{\sqrt{x}}{y^2}\right)$

c. $\log(\sqrt{x} \cdot y^2)$ ✓

d. $\log(2\sqrt{x} \cdot y)$

$\frac{1}{2}\log x = \log\sqrt{x}$; combined: $\log(\sqrt{x} \cdot y^2)$.

Part V — Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

29 [Intermediate] • Target: MCQ — Solving exponential equations (matching bases)

Solve $5^{3x} = 125$.

a. $x = \frac{1}{3}$

b. $x = 41.67$

c. $x = 3$

d. $x = 1$ ✓

$5^{3x} = 5^3 \Rightarrow 3x = 3 \Rightarrow x = 1$.

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30 [Intermediate] • Target: MCQ — Solving exponential equations using logarithms

Solve $7^x = 50$ for x, rounded to three decimal places.

a. 7.143

b. 2.010 ✓

c. 3.000

d. 1.505

$x = \frac{\ln 50}{\ln 7} \approx 2.010$.

31 [Intermediate] • Target: MCQ — Solving logarithmic equations

Solve $\log_2(x - 1) = 3$.

a. $x = 6$

b. $x = 9$ ✓

c. $x = 7$

d. $x = 8$

$x - 1 = 2^3 = 8 \Rightarrow x = 9$.

32 [Intermediate] • Target: MCQ — Natural logarithms and e

Solve $\ln(x) = 3$ for x, in terms of e.

a. $x = e^{1/3}$

b. $x = 3$

c. $x = e^3$ ✓

d. $x = 3e$

$\ln(x) = 3 \Rightarrow x = e^3$.

33 [Intermediate] • Target: MCQ — Exponential growth and decay models

Determine whether $N(t) = 80e^{-0.015t}$ models growth or decay.

a. **Decay** ✓

b. Neither

c. Growth

d. Cannot be determined

The exponent's coefficient is negative: decay.

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34 [Intermediate] • Target: MCQ — Word problems: exponential models in context

A population of bacteria doubles every 3 hours, starting at 500. Using $P(t) = 500 \cdot 2^{t/3}$, find the population after 9 hours.

a. 4,000 ✓

b. 2,000

c. 8,000

d. 1,500

$P(9) = 500 \times 2^3 = 4{,}000$.

35 [Intermediate] • Target: MCQ — Log laws: the power rule

Evaluate $4^{\frac{1}{2}}$.

a. 4

b. 5

c. 2 ✓

d. 6

$4^{\frac{1}{2}} = 2$.

36 [Intermediate] • Target: MCQ — Expanding and condensing logarithmic expressions

Evaluate $\log_4 1$.

a. 0 ✓

b. 4

c. -1

d. 1

$\log_4 1 = 0$.

37 [Intermediate] • Target: MCQ — Solving exponential equations (matching bases)

Solve $3^{x+3} = 81$.

a. 4

b. 1 ✓

c. 2

d. 0

$81 = 3^4$, so $x + 3 = 4 \Rightarrow x = 1$.

Part VI — Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)

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38 [Challenging] • Target: MCQ — Solving exponential equations using logarithms • Cross-Lesson: 1.01 Algebra & Equations

Combine $\log_2 2 + \log_2 2$ into a single logarithm and evaluate.

a. 3

b. 1

c. 4

d. 2 ✓

$\log_2 (2 \times 2) = \log_2 4 = 2$ (since $2^2 = 4$).

39 [Challenging] • Target: MCQ — Solving logarithmic equations • Cross-Lesson: 2.01 Functions & Transformations

Combine $\log_5 125 + \log_5 25$ into a single logarithm and evaluate.

a. 5 ✓

b. 6

c. 3125

d. 4

$\log_5 (125 \times 25) = \log_5 3125 = 5$ (since $5^5 = 3125$).

40 [Challenging] • Target: MCQ — Natural logarithms and e • Cross-Lesson: 3.01 Sequences & Series

Evaluate $\log_3 (27^3)$ using the power rule.

a. 12

b. 9 ✓

c. 11

d. 6

$\log_3 (27^3) = 3\log_3 27 = 3(3) = 9$.

41 [Challenging] • Target: MCQ — Exponential growth and decay models • Cross-Lesson: 4.01 Trigonometry

Solve $\log_4 (x - 1) = 4$.

a. 257 ✓

b. 255

c. 256

d. 259

$x - 1 = 4^4 = 256 \Rightarrow x = 257$.

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42 [Challenging] • Target: MCQ — Word problems: exponential models in context • Cross-Lesson: 5.01 Geometry & Coordinate Geometry

Solve $3^x = 80$ for x, rounded to three decimal places.

a. 4.090

b. 3.890

c. 3.990 ✓

d. 1.903

$x = \frac{\log 80}{\log 3} \approx \frac{1.903}{0.477} \approx 3.990.$

43 [Challenging] • Target: Free-Response Questions (FRQs): Foundational Fluency (1–7) • Cross-Lesson: 6.01 Limits & Derivatives

Solve $2^x = 70$ for x, rounded to three decimal places.

a. 1.845

b. 6.130 ✓

c. 6.030

d. 6.230

$x = \frac{\log 70}{\log 2} \approx \frac{1.845}{0.301} \approx 6.130.$

44 [Challenging] • Target: Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14) • Cross-Lesson: 7.01 Integration & Applications

Evaluate $\ln(e^2)$.

a. 1

b. e^2

c. 2 ✓

d. 3

$\ln(e^2) = 2.$

45 [Challenging] • Target: Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20) • Cross-Lesson: 9.01 Statistics & Probability

A quantity is modeled by $N(t) = 1000e^{-0.1t}$. Find $N(5)$, rounded to one decimal place.

a. 545.9

b. 606.5 ✓

c. 667.1

d. 1000

$N(5) = 1000e^{-0.5} = 1000(0.60653) \approx 606.5.$

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